



Problem-Solution Fit

Date	15 JULY 2026
Project Name	Smart Lender: Intelligent Loan Eligibility Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Banks, Financial Institutions, Credit Officers, Loan Processing Teams, Loan Applicants	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual Evaluation: Accurate but slow and labor-intensive. Basic Rule-Based Systems: Fast but less accurate. Smart Lender: Fast, accurate, scalable, and data-driven.	8. CHANNELS OF BEHAVIOR [CH] ONLINE : Smart Lender Web Application, IBM Cloud Platform. OFFLINE : Bank Branches, Loan Service Centers
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2. PROBLEMS / PAINS
+ ITS FREQUENCY [PR]

- Manual loan verification is slow.
- Difficulty identifying high-risk applicants.
- Human errors in loan approval decisions.
- Large number of applications during peak periods.

6.
CUSTOMER
LIMITATIONS

EG. BUDGET,
DEVICES

[CL]

- Limited historical credit information.
- Budget constraints.
- Need for secure data handling.
- Existing legacy banking systems.

9. PROBLEM ROOT /
CAUSE [RC]

Manual verification, inconsistent decision-making, increasing application volume, lack of intelligent prediction systems, and inefficient risk assessment.

3. TRIGGERS TO ACT
[TR]

- Increasing loan applications.
- Need for faster approval decisions.
- Reducing loan defaults and financial risk.
- Improving customer experience.

7. BEHAVIOR
+ ITS INTENSITY [BE]

- Applicants expect instant loan decisions.
- Credit officers process many applications daily.
- High demand during peak loan seasons

10. YOUR SOLUTION
[SL]

Smart Lender uses Decision Tree, Random Forest, KNN, and XGBoost models to predict loan eligibility. The best-performing XGBoost model is deployed in a Flask web application on IBM Cloud, providing real-time, accurate, and reliable loan approval predictions

4. EMOTIONS
BEFORE / AFTER [EM]

Before: Uncertainty, stress, long waiting time.

After: Confidence, faster approval, improved customer satisfaction.